MIL-C-45115(Ord)
20 June 1958
SUPERSEDING
OAC-PD-59
5 December 1956

MILITARY SPECIFICATION

CARTRIDGE, SMOKE
WP, 76MM; M361

1. SCOPE

1.1 This specification covers the assembly, marking, and packing of the M361, WP, Smoke Cartridge, for use in the 76MM Gun, M32 & T124. (See 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

75-1-367 - Cartridge, Smoke, WP, M361, W/Fuze, PD, M48A3 .05 Sec. Delay for 76MM Gun, M32 & T124. Assembly and Marking Diagram

F 8796522 - Marking Diagram and Sealing for Wood Packing Boxes

OSO 1315

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer."

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 The cartridge shall comply with all requirements specified on Drawing (Dwg.) 75-1-367, and in applicable specifications.

3.3 Propellant. - The weight of the propellant shall not differ by more than plus or minus 0.05 ounce avoirdupois from that designated by the loading authorization for the particular lot of propellant used. The contractor shall weigh and checkweigh the propellant used in the loading of each cartridge. The checkweighing shall be accomplished by an operator working independently of the original weigher, using a different scale from that used to make the first weighing.

3.4 Primers. - Each primer assembled to a cartridge case shall have the required number of flash holes as specified on the appropriate primer detail drawing.

3.5 Bullet pull. - The cartridge case shall separate from the shell at a load ranging between 5,000 pounds minimum and 11,000 pounds maximum.

3.6 Functioning.

3.6.1 Average velocity. - The difference between the velocities of any two cartridges fired in the velocity test shall not exceed 3 percent of the mean velocity of all cartridges fired in this phase of the test. The average velocity shall not differ from the prescribed velocity by more than plus or minus 2 percent after reducing results to standard conditions.

3.6.2 Premature Burst. - There shall be no premature bursts or evidence thereof (see 6.3).

3.6.3 Security. - There shall be no evidence of separation of the metal parts in the gun bore or in flight.

3.6.4 Smoke functioning. - The cartridge shall function satisfactorily (i.e., produce a smoke cloud of suitable size and density.)

3.7 Workmanship.

3.7.1 All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

3.7.2 The rotating band of each shell shall not be damaged by being nicked or burred.

3.7.3 The painting and markings shall be restored by the assembling contractor on any components furnished to him painted and marked when much painting and markings have been marred or obliterated.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General Provisions.-The quality assurance provisions contained in this section are applicable to inspection performed by or for the Government and shall be performed as prescribed below.

4.1.1 Contractor Quality Assurance System.-The contractor shall provide and maintain an effective Quality Assurance System Acceptable to the Government covering the supplies under this contract. Results of all examinations and tests performed under the Quality Assurance System will be made available to the Government. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when as a minimum, it provides the Quality Assurances required by this specification and other documents referenced in this specification and also provides for 100 percent examination of product for each listed critical defect. The contractor shall not be restricted to the inspection station nor to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute, as to whether certain procedures of the system provide equal assurance, the comperable procedure of this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Government Verification Inspection.

4.1.2.1 At the beginning of the contract, Government verification will consist of: (a) All inspections listed herein which are not reserved for sole performance by the contractor, (b) a monitoring of those inspections reserved for sole performance by the contractor, and (c) monitoring of the contractor's inspection and quality control operations. Monitoring will be defined in 4.1.2.2.

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4.1.2.2 After the contracting officer has determined that the conditions for utilization of reduced verification have been met, Government verifidation will consist of the following: Using the contractor's written Quality Assurance Prodedure, this specification and other contractual documents as a guide, the Government inspector shall monitor all quality assurance operations performed by the contractor at regular but unscheduled intervals. Monitoring consists of surviellance of the operations to determine whether the practices, methods, and prodecures of the written procedure are being properly applied. Deviations from the prescribed or agreed upon procedures, or instances of poor practices which might have an effect upon the quality of the production shall be immediately called to the attention of the contractor. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the inspector will report such instances. Failure of the contractor to promptly correct defeciencies discovered shall be cause for rejection of the lot, until corrective action has been taken to the satisfaction of the Government inspector. As an adjunct to his monitoring activities, the inspector may conduct inspection, at unscheduled, random intervals, of product from lots which have been inspected and found to be acceptable by the contractor.

4.1.2.3 Government verification of listed critical characteristics will at all times be performed as specified below.

4.1.2.3.1 Inspection will be performed by a sampling plan with an AQL of 0.015 percent. Any plan from standard Mil-Std-105 or Publication ORD-M608-11 may be used. The screening inspection normally required of the contractor in conjunction with plans from ORD-M608-11 or in case of rejection by the Mil-Std-105 plans will be in addition to the contractor inspection performed prior to presentation.

4.2 Acceptance inspection.

4.2.1 Lot formation. - A lot shall consist of cartridge assemblies produced by one manufacturer, in one unchanged process in accordance with the same drawing, same drawing revision, same specification, specification revision, under conditions of continous production free from interruptions other than those due to the end of a shift, day, or workweek. (See 6.2) In addition, the lot shall have the following limitations:

a. Cartridge cases limited to one manufacturer, manufactured under the same drawing revision and same specification and specification revision,,and drawing.

b. Propellant from not more than one lot.

c. Loaded primers from not more than two lots. These lots should be consecutive lots, if possible (Not a requirement.)

d. Loaded and fuzed shell from not more than one lot.

4.2.1.1 Functioning. - Fifteen cartridges will be selected by the inspector for the test specified in 4.4.

4.2.1.2 Bullet Pull. - Ten cartridges will be selected by the inspector for the test specified in 4.3.1.

4.2.1.3 Chamber Gage. - After the cartridge has been assembled, the contractor shall gage each cartridge in a profile and Alignment gage (chamber gage) pertaining to the applicable cartridge. If the cartridge fails to enter the chamber gage freely, it shall be rejected, subject to correction and resubmission for test. Drawings of the Government profile and alignment gage will be furnished to the contractor for his information upon application to the inspector.

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4.2.1.4 Primers.-The contractor shall visually inspect each primer for one or more holes missing, prior to insertion into the cartridge case.

4.2.2 Examination.-Inspection will be performed as specified in specification MIL-G-2550 and as indicated below. Samples and acceptance numbers will be related in accordance with Standard MIL-STD-105 or Publication ORD-M608-11, as appropriate.

4.2.3 Tests.-The tests specified in paragraph 4.3 shall be conducted by contractor as specified in MIL-G-2550 in accordance with the methods specified in paragraph 4.3. Sampling and acceptance criteria shall be as specified with the test description.

4.2.4 Classification of defects.-Inspection of the cartridge will be performed as follows:

4.2.4.1 Shell filling assembly, prior to insertion of burster initiator (see Dwg. 75-14-694, covering a detail of Dwg. 75-1-367).

Categories and defects	Method of inspection
Critical	
a. Burster pad or pads missing.	Visual
Major AQL 0.25 percent	
1. Burster charge assembly missing.	Visual

4.2.4.2 Shell filling assembly (see Dwg. 75-14-694 covering a detail of Dwg. 75-1-367.)

Categories and defects	Method of inspection
Critical	
a. Zone weight off two or more zones.	Balance
b. Evidence of leakage.	Visual
Major AQL 0.40 percent	
1. Zone weight off one zone.	Balance
2. Depth of fuze cavity, min.	Gage
3. Detonator missing.	Visual
4. Burster initiator missing.	Visual
Minor AQL 1.50 percent	
41. Retainer missing or loose.	Visual-Manual
42. Threads not coated (when applicable).	Visual
43. Foreign matter on exterior.	Visual

4.2.4.3 Assembly prior to insertion of loaded shell (see Dwg. 75-1-367)

Categories and defects	Method of inspection
Critical	
a. Propellant weight beyond assessed weight tolerance by 1% or more.	Balance
b. One or more primer flash holes missing.	Visual

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Major AQL 0.40 percent

1. Propellant weight beyond assessed weight tolerance by less than 1%. . . . . .Balance
2. Primer loose. . . . . .Torque Test (see 4.2.5.1)
3. Primer body detached or missing. . . . . Visual

Minor AQL 1.50 percent

41. Foreign matter in charge. . . . . .Visual
42. Foreign matter on interior of case. . . . . .Visual
43. Distance wadding missing or damaged. . . . . Visual

4.2.4.4 Cartridge assembly (see Dwg. 75-1-367)

Categories and Defects	Method of Inspection
Critical	
a. Propellant charge missing or obviously light. . . .	.Balance
b. Primer above flush.	.Visual-Gage
c. Zone stenciling incorrect by two or more zones. . .	.Visual
Major AQL 0.40 percent	
1. Incorrect type fuze.	Visual
2. Primer missing.	.Visual
3. Zone stenciling off one zone.	.Visual
4. Round fails to chamber gage freely.	.Gage
Minor AQL 4.00 percent	
41. Fuze loose.	.Gage
42. Excessive gap between cartridge case and shoulder of rotating band.	Gage-Visual
43. Excessive gap between fuze and projectile.	Visual (see 4.2.5.2)
44. Primer obviously damaged.	.Visual
45. Crimp of cartridge case torn.	.Visual
46. Cartridge case split or cracked.	Visual
47. Foreign matter on exterior of round.	Visual
48. Marking misleading or unidentifiable.	.Visual
49. Rotating band with a cut or embedded foreign matter.	.Visual(See 4.2.5.3)
50. Bare spot in excessive of 1/4 sq. inch.	.Visual
51. Paint on rotating band bearing surfaces.	Visual

4.2.4.5 Cartridge taken from unsealed fiber container (see Dwg. 76-1-1320), covering a detail of Dwg. 75-12367)

Categories and defects	Method of Inspection
Major AQL 0.25 percent	
1. Packing stop missing.	.Visual
Minor AQL 1.50 percent	
41. Round cannot be removed by hand.	Manual
42. Glue or asphalt on round.	.Visual
43. Improper assembly.	Visual

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4.2.4.6 Sealed fiber container (see Dwg. 76-1-1320, covering a detail of Dwg. 75-1-367)

	Categories and defects	Method of Inspection
Major	AQL 0.40 percent	
1.	A cut completely through all impregnated asphalt layers.	Visual
2.	Metal end loose or distorted.	Visual
3.	Tape incomplete or badly wrinkled.	Visual(see 4.2.5.4)
Minor	AQL 1.50 percent	
41.	Gap between cover and body of container, max.	Visual(see 4.2.5.5)
42.	Tear tab length, min.	Gage
43.	Contents loose.	Manual
44.	Marking misleading or unidentifiable.	Visual

4.2.4.7 Sealed wood box (see Dwg. 76-1-1321, covering a detail dwg. 75-1-367)

	Categories and defects	Method of Inspection
Major	AQL 0.40 percent	
1.	FSN missing, incorrect or unidentifiable.	Visual
2.	Strapping missing, broken, or loose.	Visual
3.	Hardware missing, broken, or loose.	Visual
4.	Board broken or split.	Visual
Minor	AQL 1.50 percent	
41.	Car seal missing, unsealed, or improperly positioned.	Visual
42.	Contents loose.	Manual
43.	Handle missing or insecure.	Visual-Manual
44.	Marking misleading or unidentifiable.	Visual

4.2.5 Defect standards

4.2.5.1 Primers shall be tested by the application of a 100 inch pound torque in the assembly direction. If the primer moves in excess of 1/8 inch, the assembly will be classed as defective.

4.2.5.2 Any assembly having a gap extending 360° around the circumference shall be classed defective. However, if contact is made at any one point, then those with a gap in excess of .020 inch shall be classed defective.

4.2.5.3 The shell shall be classed defective if the rotating band contains any longitudinal cut deeper than 1/4 the distance between the surface of the band and the surface of the shell, for the full width of the band, or if it contains foreign matter embedded to a similar depth and width.

4.2.5.4 When the inner layer of tape is not in contact with the surface of the container for at least 1/2 inch on each side of the joint around the entire circumference, the container shall be classed defective.

4.2.5.5 The inspection for an excessive gap between the cover and body of the container will be performed by feeling with the fingernail along the tape. If it is suspected that an excessive gap is present, the tape will be unwound until enough of the joint is visible for determination.

4.2.6 Inspection equipment.

4.2.6.1 Government Verification Inspection.-Government verification inspection will be accomplished with the use of gages and equipment that has been certified, by the design specified in the "List of Inspection equipment, Numbers Cadt, Smk, WP, M361 (C.R.)-GL-75-1-367; Shell, Smk, WP, M361 (ldg.)-GL-75-1-14-694A; Fiber container - GL-76-1-1320.

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4.2.6.2 Contractor inspection.-Unless otherwise specified herein, contractor inspection will be accomplished with the use of gages and equipment designated to perform, to the required accuracy, the inspections and controls delineated in the contractor's written description specified in 4.1.1. In inspecting the contractor's inspection equipment the Government inspector will determine that the contractor has available and utilized correctly, gaging, measuring and test equipment of required accuracy and that the instruments are of proper type and range to make measurements of the required accuracy. The contractor shall have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the supplier and made available for review by the government. The calibration of gages, standards, and instruments will be periodically checked by authorized Ordnance personnel.

4.3 Tests

4.3.1 Bullet pull.-Ten sample cartridges shall be tested to determine the security of the crimp of the cartridge case to the shell, by applying a load in the axial direction, rate of travel of the testing machine head to be 0.100 to 0.125 inch per minute inclusive. If one or more of the samples fail to meet the requirements of 3.4, the lot shall be rejected.

4.4 Functioning.-The following test will be conducted by the Government at a proving ground as specified by the procuring activity.

4.4.1 Procedure.

4.4.1.1 Average velocity.-Fire five of the sample cartridges, with fuzes set at "SQ", in a gun for which the cartridge is standard for impact on firm ground, and measure the velocities.

4.4.1.2 Safety.-Disassemble five of the sample cartridges and reassemble with propellant adjusted to give 112 percent of the average pressure given by the propellant of the particular ammunition lot. (This value of pressure appears on the data card of each ammunition lot). Fire in a gun for which the cartridge is standard, at any convenient gun elevation, with fuzes set at "SQ", and make observations for premature bursts and separation of metal parts in the gun bore or in flight before impact.

4.4.1.3 Smoke.-Fire five of the sample cartridges, with fuzes set at "SQ", in a gun for which the cartridge is standard, for impact against a one-inch wood screen placed not less than 400 feet from the muzzle of the weapon. Observe for functioning of the shell, and for size and character of the smoke cloud. Functioning of the shell on or not more than five feet beyond the screen shall be considered satisfactory.

4.4.2 Retest and rejection.-A retest may be made, if requested by the contractor, using double the number of samples used in the original test, provided that:

a. No shell failed to meet the requirements of 3.6.2 and 3.6.3.

b. Not more than one cartridge fails to produce smoke cloud of suitable size and density,

c. The difference between the velocities of any two cartridges fired in the velocity test exceeds 3 percent but does not exceed 4 percent of the mean velocity of all cartridges fired in this phase of the test and/or,

d. The average velocity differs from the specified velocity by more than plus or minus 2 percent, but not more than plus or minus 3 percent. The lot shall be rejected, not subject to further test if, on retest:

a. One or more shell fail to meet the requirements of 3.6.2 and 3.6.3,

b. One or more shell fails to produce a smoke cloud of suitable

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size and density,

c. The cartridges fired, in the average velocity phase fail to meet the requirements of 3.6.1.

4.5 Acceptance.-If the requirements of the above inspections and test have been met, and if the inspector is assured that all other requirements have been met, the lot will be accepted.

4.5.1 At the beginning of the contract and as long as the Government inspector continues to perform the verification specified in 4.1.2.1 acceptance will be based on the results of the Government verification specified in 4.1.2.1.

4.5.2 After reduced Government verification specified in 4.1.2.2 has been initiated, acceptance will be based on the results of the contractor's inspection and quality control and the inspections reserved solely for performance by the Government and the results of the monitoring by the Government of the contractor's quality assurance system.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.2 Level A.-Packaging shall be as specified on Dwg. 76-1-1320.

5.2 Packing.

5.2.1 Level A.-Packing shall be as specified on Dwg. 76-1-1321.

5.3 Marking.

5.3.1 Level A.-Marking shall be as specified on Dwg. F8796522.

5.4 Data Cards.-Data card information shall be as specified in Specification MIL-C-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Cartridge.-A cartridge as referred to herein means the assembly of all components of ammunition into one unit necessary for firing the shell from a gun; for example: cartridge case, loaded percussion primer, propellant, and loaded shell, all properly assembled.

6.3 Premature burst.-A "Premature Burst" is defined as a functioning of the shell in the gun bore or in flight before impact.

6.4 When the contractor for the cartridge is also the contractor for one or more of the components thereof, the functioning tests of components may be combined with the functioning tests of the cartridge to save expense, upon agreement between the Government and the contractor to that effect. In case the cartridge specification does not cover all of the functioning tests specified for the components the additional tests prescribed in the component specifications will be conducted.

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